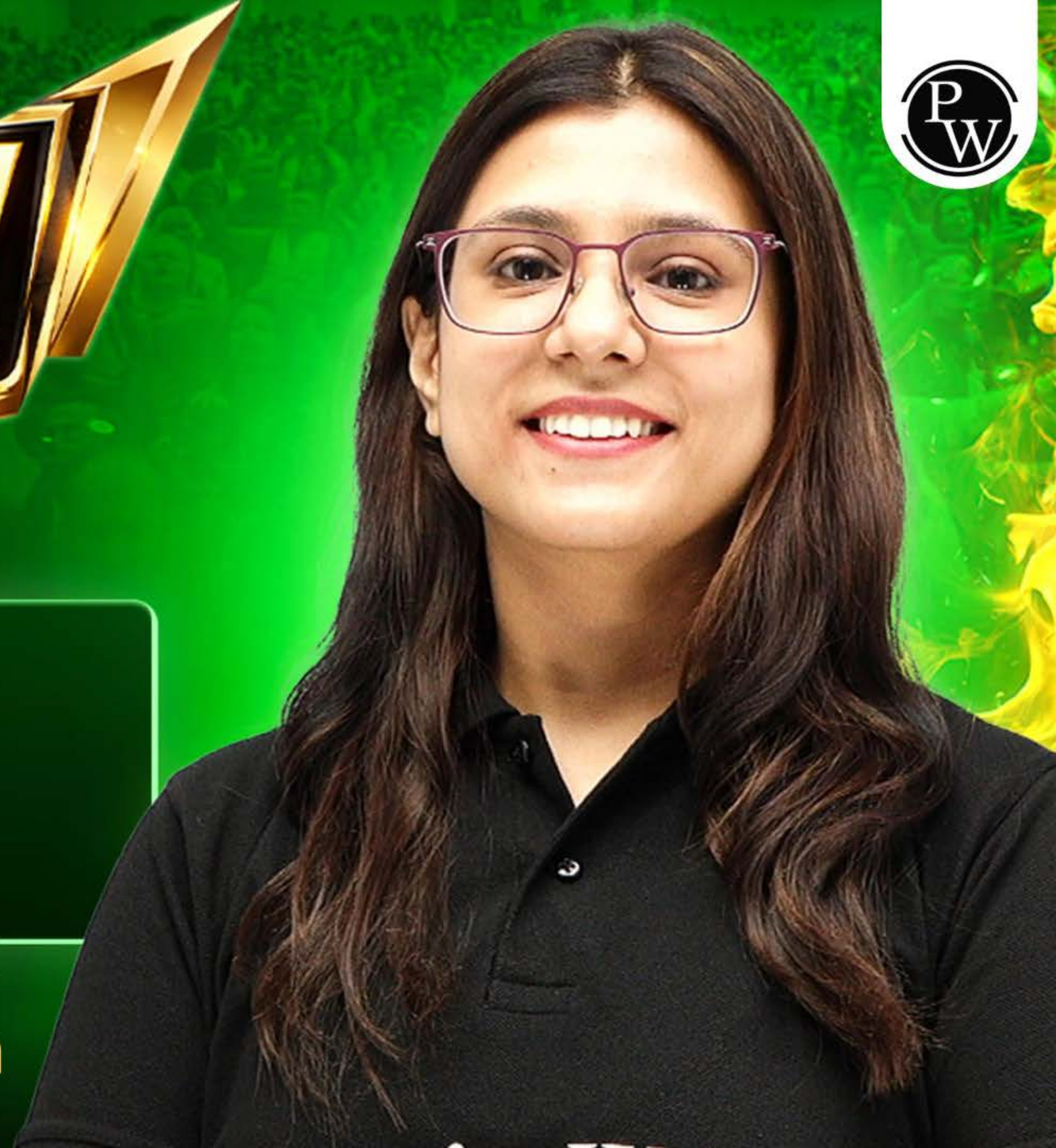
VICTORY 2027

Chemical coordination in plants

LECTURE-02 | BIOLOGY

By– Upasana Ma'am



TOPICS TO BE COVERED

Plant Hormones and Tropic Movements


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

Stimuli and Response

- **Stimuli** is a change in either external or internal environment of an organism.
- **Response** is the resulting actions or movements caused by stimuli.

- Plants respond to stimuli by producing chemical compounds called "hormones" that work as messengers.
- The term hormone was first used by William Bayliss and Ernest Starling in 1902.
- Plants are sensitive to light and respond to the direction of light. Also sensitive to water, temperature, gravity, etc.



Plant Hormones

1. Auxins :



- The term Auxin was first coined by F.W. Went in 1928 from the Greek word auxin which means 'to grow'.
- This is more effective at low concentrations.
- It promotes cell elongation, which increases cell wall.
- Plant cell wall get loosened like rubber when put under tension, rate of cell elongation is equal to auxin concentration.



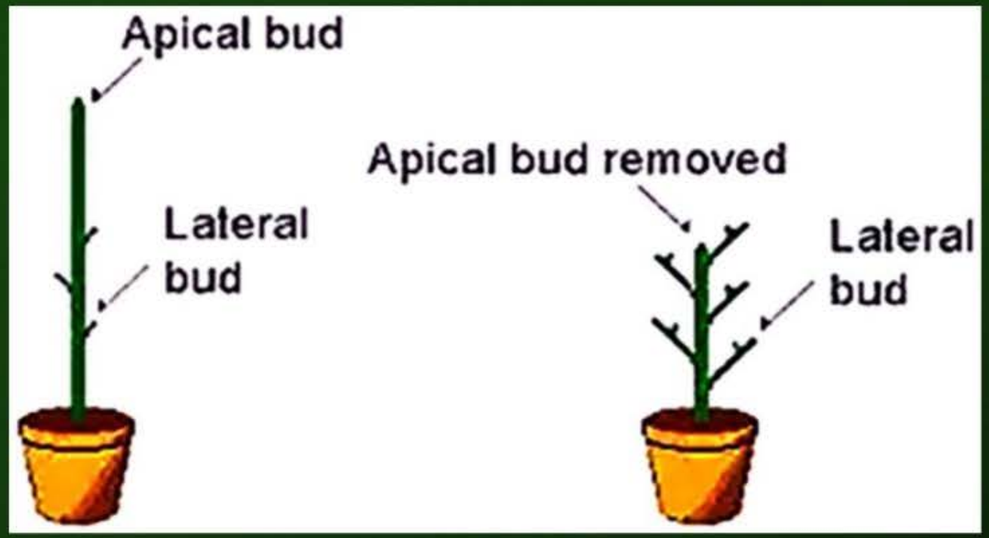
Plant Hormones

- First growth hormone discovered in plants.
- Are powerful growth-stimulants and also effective at extremely low concentration.
- IAA (Indole-3-acetic acid) main natural auxin in plants.
- Present in shoot apex, root apex, lateral meristems, etc.

Plant Hormones

❑ Apical Dominance

- The phenomenon of the suppression of the growth of lateral buds by apical buds is called Apical Dominance.



Plant Hormones

2. Gibberellins :

- Different forms of gibberellins from different plants – GA_1, GA_2, GA_3 and so on.
- Gibberellic acid (GA_3) is most studied.
- Promote growth of internodes by cell elongation.
- Present in meristematic region like stem-apex, root-apex, buds, seeds, etc.
- Breaks seed dormancy and initiate germination.
- Promote fruit growth and are capable of inducing parthenocarpy.
- Delay senescence (ageing).





Plant Hormones

- Causes elongation of stem

Gibberellin



Plant Hormones

cell division

3. Cytokinin :



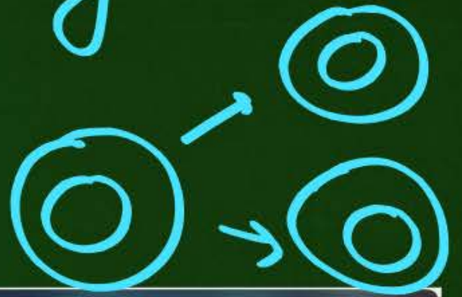
- Cytokinin have been discovered comparatively recently 1950s by Skoog and Miller.
- Cytokinin help in Cell division.



Auxin, Gibberlin.

↓

cell elongation



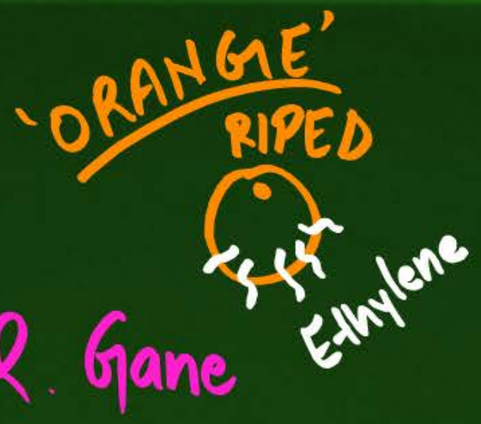


Plant Hormones

3. Cytokinin :

- Stimulate plant growth by cell division as against auxin and gibberellins which stimulate growth by cell elongation.
- Widely distributed in plants.
- Produced in root tips and transported through xylem.
- Found in germinating seeds, developing fruits, embryo etc.
- Cytokinin inhibit apical dominance.
- They promote chlorophyll synthesis in chloroplasts and delay leaf senescence.

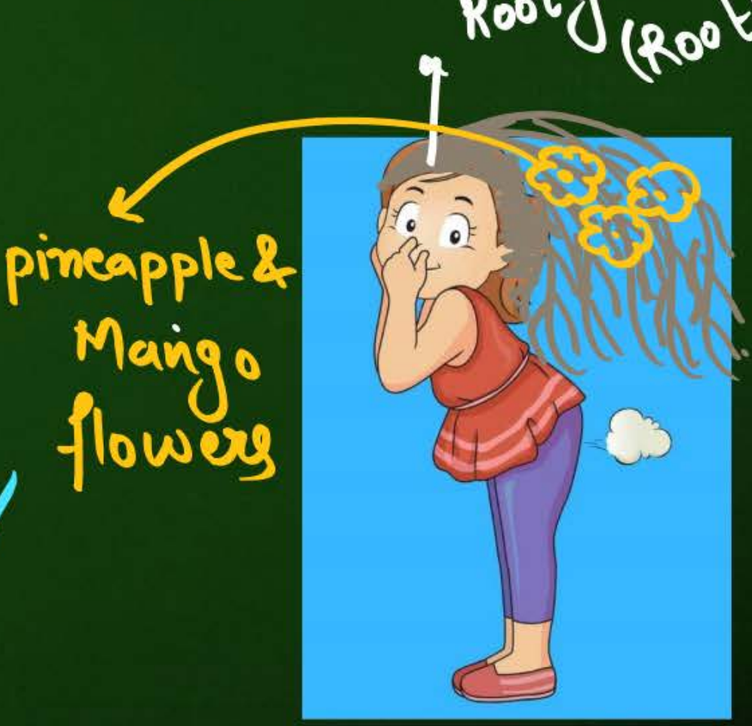
Plant Hormones





4. Ethylene

- Only hormone which is a gas at ordinary temperature.
- Site of synthesis and site of action are not different.
- Gas causing ripening of fruits.
- Produced in higher plants and fungi.
- More ethylene is produced in meristematic tissues.



Plant Hormones

4. Ethylene

Reduction in stem elongation and acceleration of senescence.

- Ripening of fruits.
- Initiating germination in peanut seeds.
- Sprouting of potato tuber.
- Promoting root growth and root hair formation.
- Inducing flowering in mango. and pineapple.

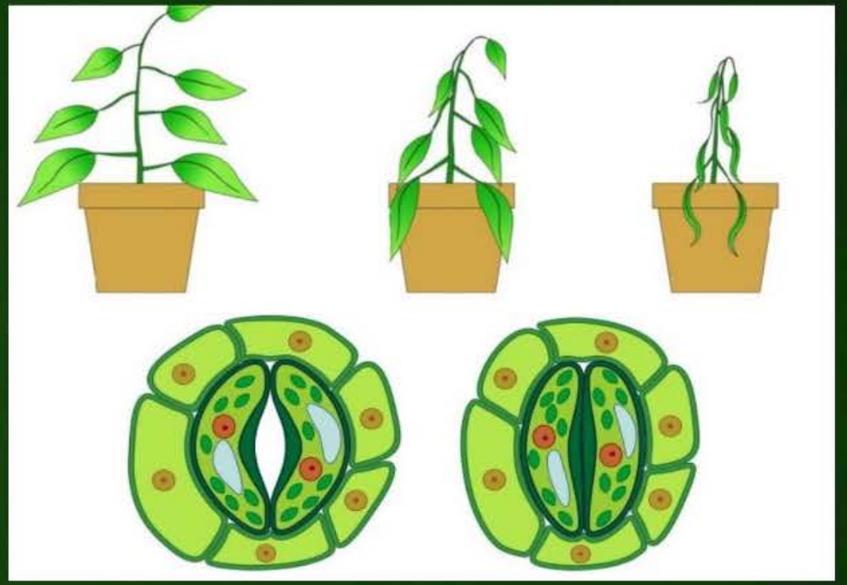


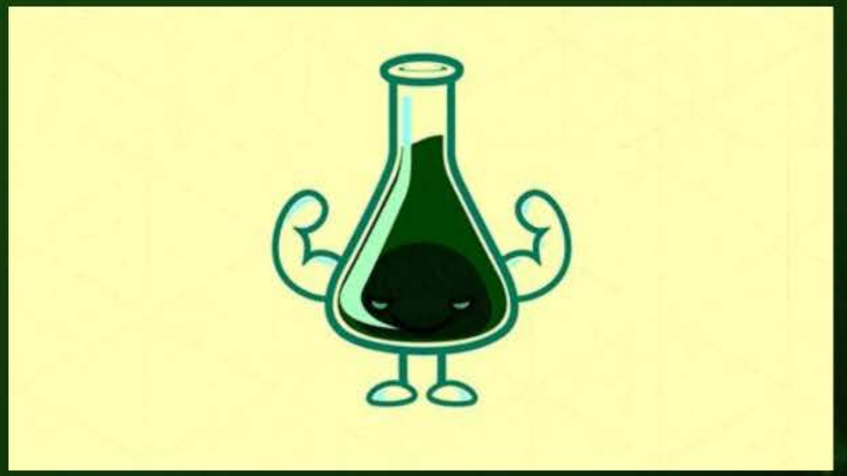
Plant Hormones

5. Abscisic acid (ABA)

inhibiting, suppress

- Growth-retarding hormone.
- Found in angiosperms, gymnosperms, pteridophytes and some mosses.
- Found in chloroplasts of leaves.
- Fruits and seeds contain highest amount of ABA.
- Also known as stress hormone.



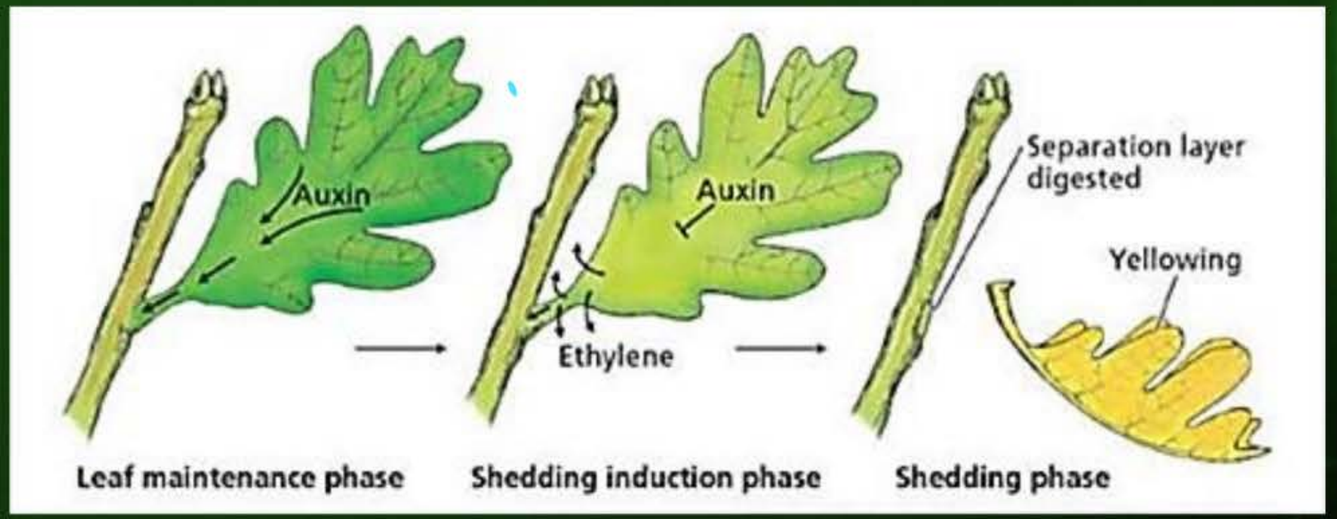


Plant Hormones



5. Abscisic acid (ABA)

- Slows down plant metabolism.
- It induces seed dormancy and helps seeds to withstand desiccation (extreme dryness).
- It accelerates senescence and abscission of leaves, buds, flowers and fruits.
- Stimulates the closure of stomata in the epidermis and increases the tolerance of plants to stresses.



Plant hormone	Site of synthesis	Functions
1. Auxins	• Shoot apical meristems and young leaves	• Promote cell elongation Suppress the growth of lateral buds Delay fall of leaves Induce formation of parthenocarpic fruits.
2. Gibberellins	• Meristem of apical buds and roots	• Help in stem elongation Break dormancy of seeds and buds Delay senescence Induce parthenocarpy
3. Cytokinin	• Roots and transported to other organs	• Stimulate cell division and cell enlargement Prevent ageing of plant parts Inhibit apical dominance
4. Ethylene	• Senescent leaves and flowers, germinating seeds and ripening fruits.	• Induces fruit ripening Promote senescence
5. Abscisic acid	• Green fruits and seeds at the beginning of wintering periods.	• Induces dormancy of buds and seeds Inhibits seed germination and development Stimulates closing of stomata

Question

Indole-3-acetic acid is the most common naturally occurring plant hormone of _____ class.

A Gibberellin

B Auxin

C Ethylene

D Cytokinin

Question

Which of these is not a function of auxin?

A inducing callus formation

B inducing dormancy

C enhancing cell division

D maintaining apical dominance

Question

Which of the following hormones is not a growth inhibitor?

A Dormin

B Abscisic acid

C Ethylene

D IAA

Question

Which of the following is the application of synthetic auxin?

A Killing the weeds.

B Ripening the fruits.

C Preventing the elongation of the internodes.

D None of these

Question

Lateral buds are not able to grow in the presence of an apical bud during the apical dominance phase. This is because of

A. The amount of cytokinin in the lateral buds is more.

B. The amount of cytokinin in the lateral buds is Less.

C. The amount of Auxin in the lateral buds is more.

D. The amount of Auxin in the lateral buds is Less

Question

What distinguishes movements in plants from movements in ammals?

A. Animals depend on light, plants do not.

B. Plant movement is due to nerves, animal movement is not.

C. Plants synthesize their own food, while animals depend on others.

D. Plants move continuously, animals do not.

Question

What is parthenocarpy and which hormone helps induce it?

A. Seed germination – Cytokinins

B. Fruit formation without fertilization – Auxins

C. Leaf fall – Gibberellins

D. Bud growth – Cytokinins

Question

What is a stimulus in a living organism?

A. A chemical change within the body

B. A result of hormone action

C. A change in the internal or external environment

D. A voluntary body movement

SECTION B (40 Marks)

(Attempt any four questions from this Section.)

Question 3

(i) Write the overall chemical equation for photosynthesis. [1]

(ii) Ramesh is a farmer who has a vineyard and cultivates grapes. To improve the shape of the grapes and make them elongated he sprays a particular *phytohormone* at the time of fruiting. [2]



Gibberlins

(a) Name the *phytohormone* that Ramesh sprays on the plant at the time of fruiting.

(b) These grapes are said to be seedless. What is the technical term for the formation of such fruits? Parthenocarpy

Q. Which of following hormone is gaseous at room temperature?

A) Auxins

B) Gibberllins

C) Ethylene ✓

D) IAA

Tropic movements in plants

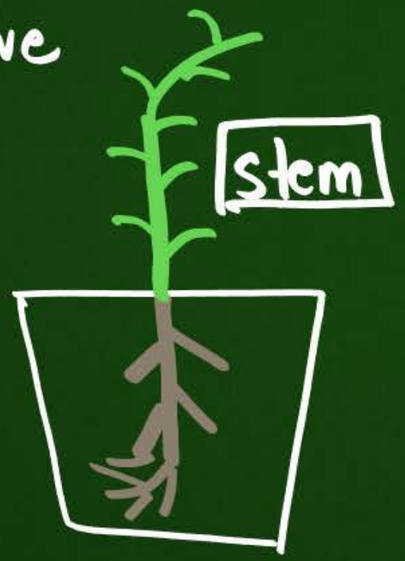
- The root must grow downwards into soil which will provide support, water and minerals for the plants.
- Shoot must grow up towards the light source so that developing leaflets could carry out photosynthesis in order to produce food for the growing plant.

The movement in parts of plant which are directional and growth related.

Directional & Growth Related

Stem — Positive → Phototropism (light)

Roots — Negative





(Sunlight) - Photo
external stimulus

[Geotropism]

Roots ——

A) Positive Geotropism

B) Negative ,,

5 pm extra class

Thank You